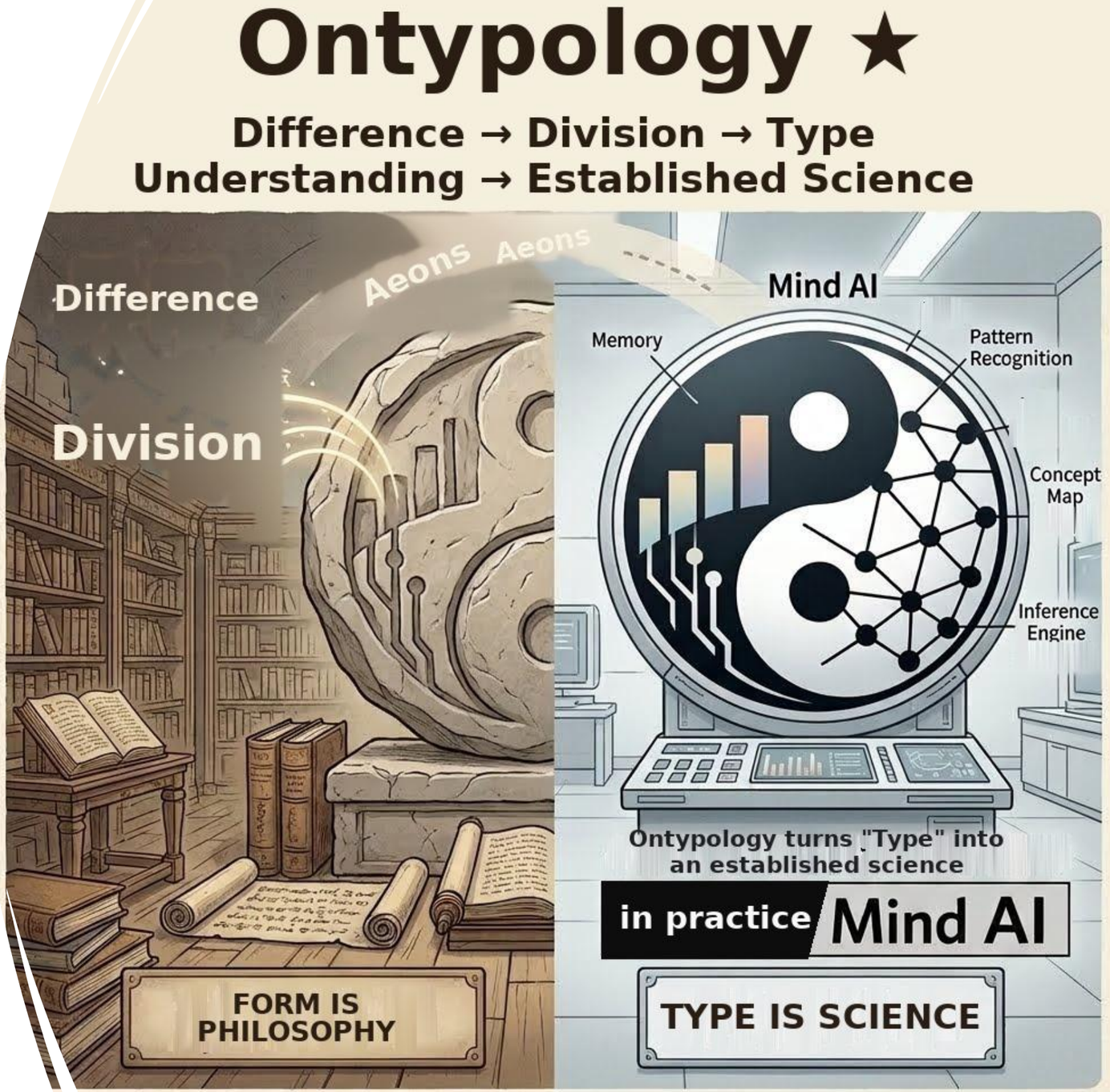
Ontypology

- From difference to division, from division to type, from type to understanding — the brain has walked this path for eons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.



Unit Type Set Logic: The Three-Dimensional Description and Judgment Architecture of Type-Driven Design

Understanding a Type = understanding what it looks like, where it is, what it is composed of, and how its parts depend on one another.

Type = { Components, (Contents), [Properties] }



Three conventions for reading this book

1. Convention One

The English for Type is always written Type (e.g. SpecType, ModuleType). Never render it as Form.

2. Convention Two

Each bracket has its own duty: { } Composition Set, [ ] Abstract Set, ( ) Container Set.

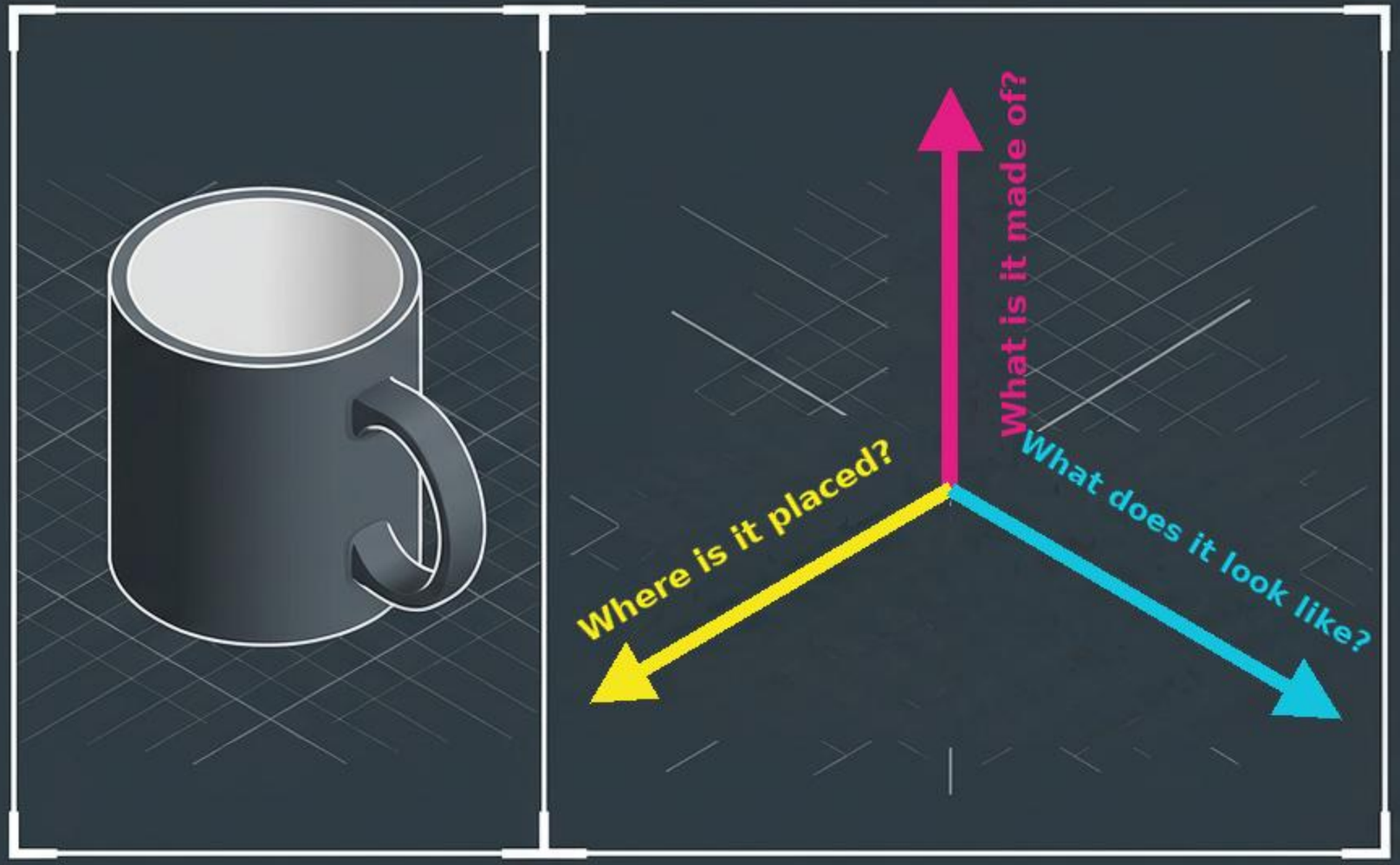
3. Convention Three

One cup runs through the whole book as the truest specimen of a Unit Type.

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The three facets that fully describe a Unit Type ("Type")

All three must be present for a Unit Type to be fully described; each facet answers one kind of question, and none can replace another.



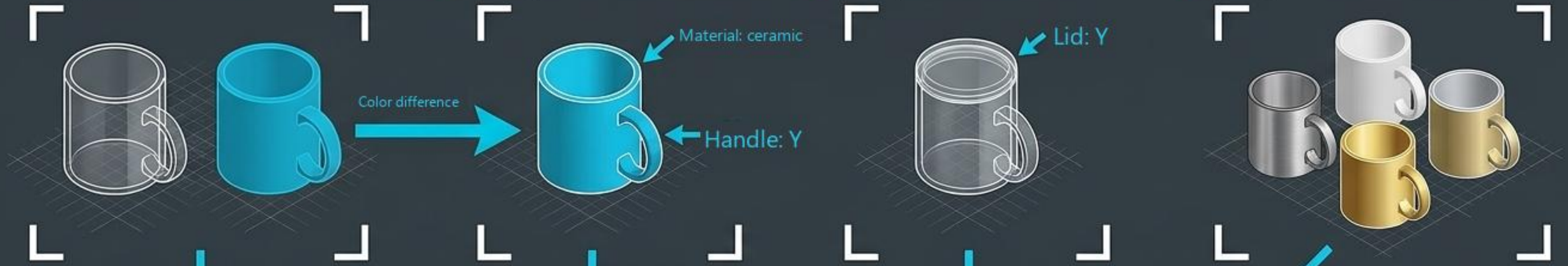
Color-coded nodes	Question	Notation	Concept
X-axis (Cyan)	What does it look like?	[ ]	Abstract Set (Property)
Y-axis (Magenta)	What is it made of?	{ }	Composition Set (BOM)
Z-axis (Yellow)	Where is it placed?	( )	Container Set (Container)

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Chapter 1 X Axis: Property [ ] - The Step-by-Step Emergence of Dimensions and Gradations

Property = a question that can be asked (dimension); property value = a concrete answer (gradation).

Step 1: Two cups → Step 2: A third cup → Step 3: A fourth cup → Step 4: All later cups



Differences fit existing properties: add gradations only, no new axis.

Property-Value Master Table

Property (dimension)	Property value (gradation)
Color	[transparent, blue, gold, silver, white...]
Material	[glass, ceramic, stainless steel, paper...]
Handle	[Y / N]
Lid	[Y / N]

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Abstraction: from a closed list to an open rule

Type Logic's “abstraction” supplies what a mathematical set most lacks: a rule that keeps generating new members.

Closed Set { }

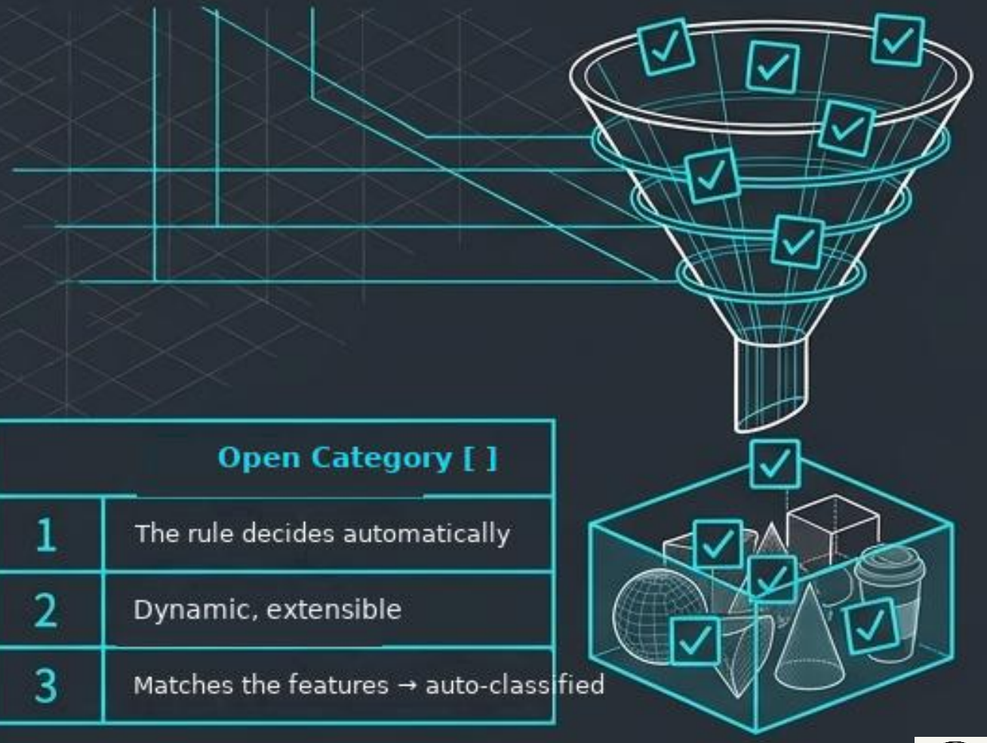


	Closed Set { }
1	Manually listed
2	Statically fixed
3	Not on the list → not a member

Open Category [ ]



Intersection criterion ∩(T): object x belongs to T ⇔ x ∈ F(T), $f_i(x)$ = true

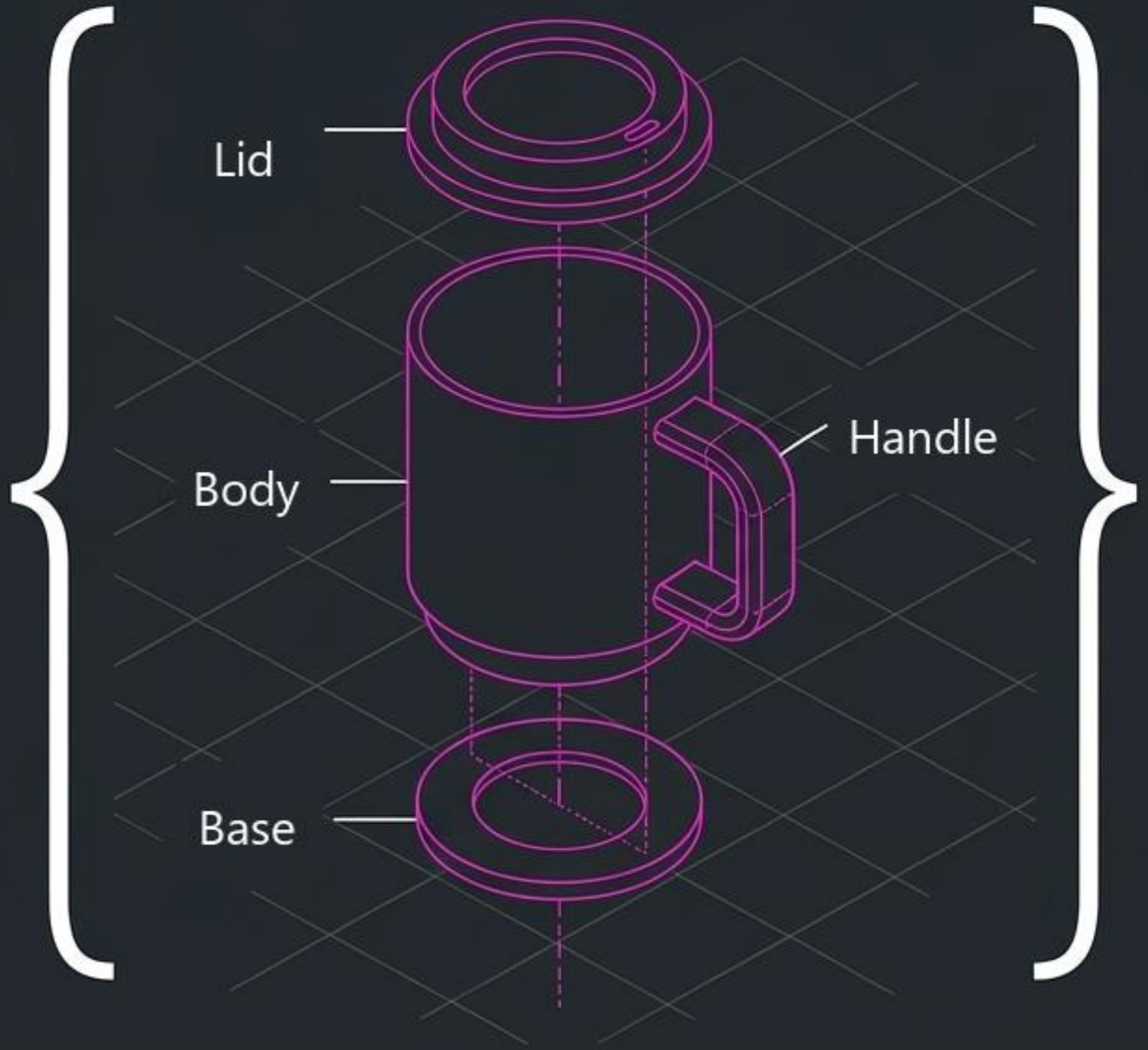


	Open Category [ ]
1	The rule decides automatically
2	Dynamic, extensible
3	Matches the features → auto-classified

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Chapter 2 Y Axis: Bill of Materials BOM { } - The Structural Blueprint

Open it up: see which smaller things compose it. This relies not on comparison but on disassembly.



The Self-Recognition Rule



Once taken off, does it still recognize itself?

Feature	State after removal	Assignment
Handle / lid	still a handle / a lid	Counts into BOM { } (Y axis)
Color / material	removing color leaves no portion of color	Goes to Property [ ] (X axis)

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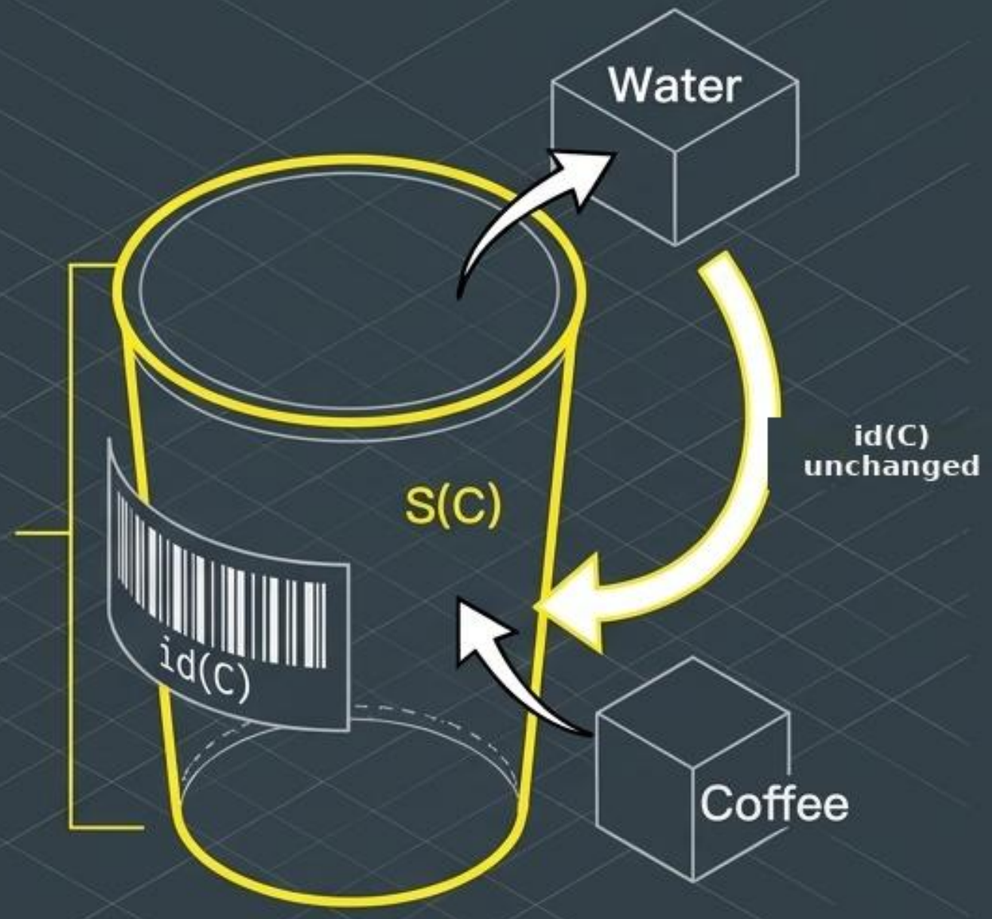
Three-Level Classification of Parts: the Remove Criterion



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Chapter 3 / Z-axis: Spatial Container ( ) — an identity that does not change with its contents

Container = a set + an independent identity that does not change with its elements



3 Elements

1. Boundary

separates inside from outside (may be closed or open)

2. Internal Space

the region that holds contents (may be empty, or hold several things)

3. Independent Identity

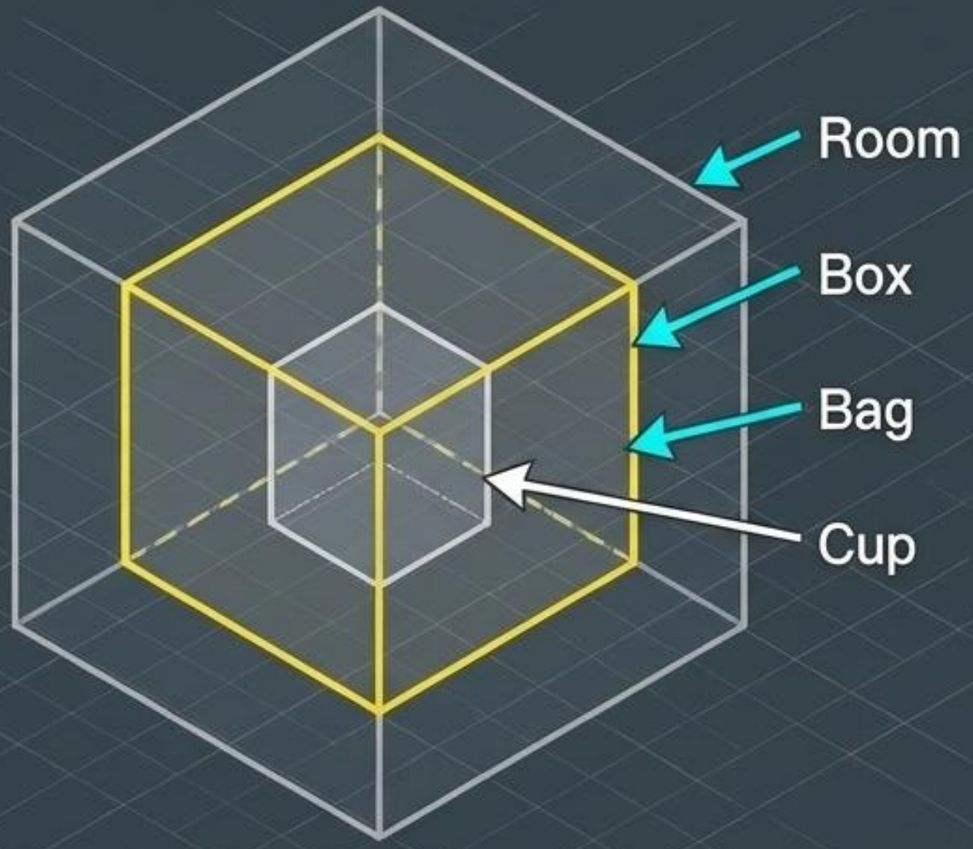
$$C = (id(C), S(C))$$

Axiom of invariance: for $e_1, e_2 \in S(C)$, replacing the internal elements does not change id(C)

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The hierarchy and nesting of containers

A cup and its water are entities at different levels. If you use a plain mathematical set { cup, water }, the main hierarchy is flattened away.



Identity

Primary key (PK): id(C) — who I am

Contents

Foreign key (FK): what is held inside

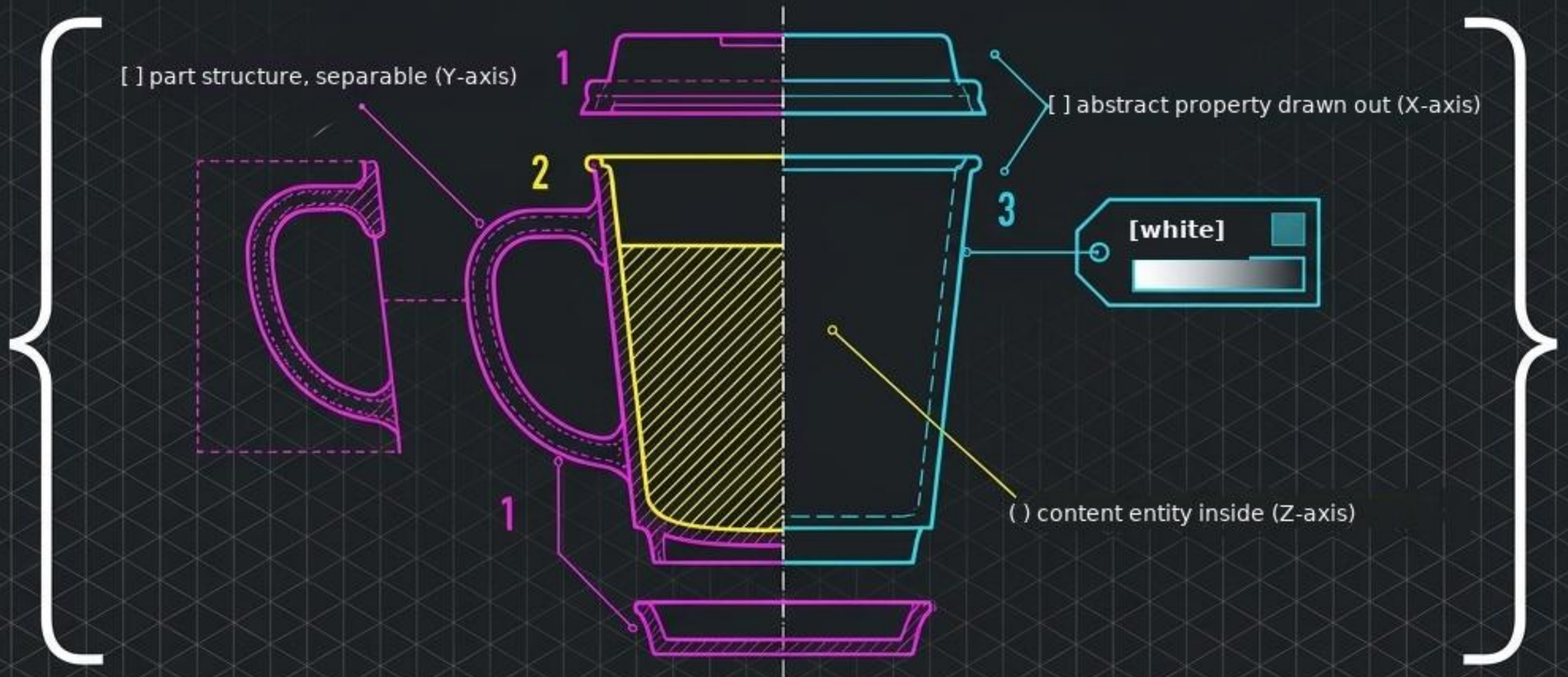
Establish the identity first, then discuss the contents.

Each level of container stands on its own; in nested layers every level obeys the same three elements — boundary, space, independent identity.

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Chapter 4 — The complete notation of a Unit Type

Unit Type = Type Set = coffee cup: { handle, base, lid, body, (coffee), [white] }



• A Unit Type is one extremely stable structure of “parts + contents + appearance”.

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How the three value-adds stack

A mathematical set answers only the most primitive question, “membership”, and omits three layers of real-world information. Type Set adds them back systematically.

The limit of a mathematical set	Type Set value-adds
{ body, lid, water, white } → just a list of elements; every level is flattened.	Value-add 1: Container ( ) — restores “the independence of identity” → the set gains a “self” that does not change.
	Value-add 2: Composition { } — restores “structure between elements” → no longer a mere list; structural dependency appears.
	Value-add 3: Property [ ] — restores “the rule that generates” ➔ A hard-coded list is upgraded into something that can be judged dynamically.

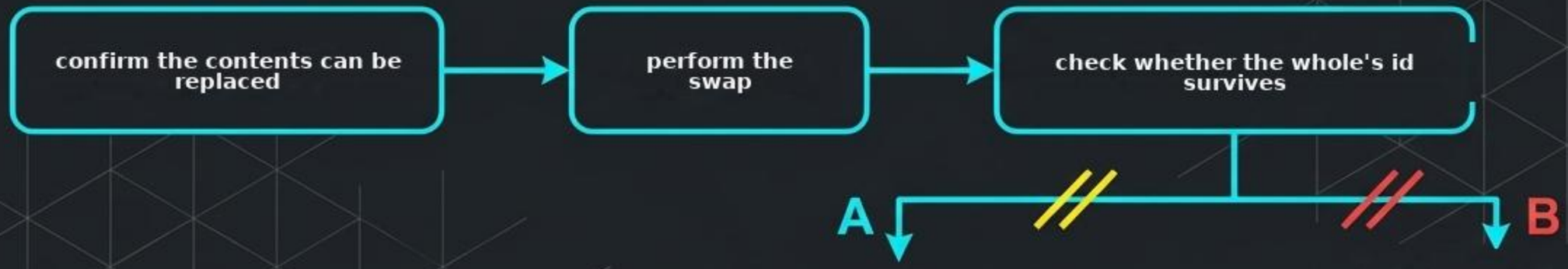
The everyday sentence “there is water in the cup” actually carries three deep layers at once: set relation, container value-add, and the replaceability of contents.

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Chapter 5

Type Logic — three-layer dynamic judgment (1/3: container layer)

Facing “an assembled whole that holds something”: is its identity independent of its contents?



swap the textbook for a pencil case
→ still the same backpack
→ container logic ( ) applies

set { apple, banana }
apple → orange
→ becomes another set
→ a pure set, not a container

Discrimination: the books on a shelf
the physical shelf = a ( ) container relation.
the books on the shelf = a { } composition relation (swap a book and the composition changes)
the judgment depends on which “whole” you are looking at.

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Dynamic judgment (2/3: composition layer & 3/3: abstraction layer)

Composition layer (Y-axis):
remove a part — is it still the same thing?

Test case 1: remove a car's steering wheel
→ cannot steer, still recognizable as a car

→ function-degraded part

Test case 2: contrast — remove the engine
→ cannot move, loses its identity as a car

→ essential part

Abstraction layer (X-axis):
does the new object satisfy the category rule ∩(T)?

Physical Test	sports bottle (holds liquid ✓, drink by hand ✓) ∈ [cup]
	thermos (must pour out to drink ✗) ∉ [cup]
Digital Test	F = { receive, generate, maintain context }
Digital Test Outcome	FAQ bot (cannot hold context ✗) ∉ [dialogue system]
	ChatGPT (satisfies all) ∈ [dialogue system]

The precision of a category boundary depends entirely on its feature set F(T) and on the completeness of its definition.

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Three-Layer Joint Judgment - An Indispensable Cognitive Filter

Layer	Test object: smartphone	Test object: Tesla EV
Layer 1: Container judgment ( )	Apps swapped, phone identity unchanged (✓)	Different passengers, body identity unchanged (✓)
Layer 2: Composition judgment { }	Mainboard (essential), speaker (downgraded) (✓)	Battery pack replaces the engine as (essential). Insight: BOM cares about structural role, not implementation (✓)
Layer 3: Abstraction judgment [ ]	Satisfies F(communicate, compute, carry) (✓)	Satisfies F(road driving, carry, powertrain) (✓)

All three layers are indispensable: by container alone a tablet is a phone; by abstraction alone a walkie-talkie is a phone.
Only passing all three precisely defines a Type.

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Summary: the complete map of Type, and the self-check

1. Appearance (X-axis / Property)

Which properties are forced out by which differences?
What is the abstract intersection criterion $\cap(T)$?

2. Composition (Y-axis / BOM)

If a part is disassembled, is it still itself?

Apply to ANY System, Product, or Object

3. Dependency (Y-axis / Logic)

After the remove(p_i, M) test, which dependency level does a part fall into (essential / degraded / non-essential)?

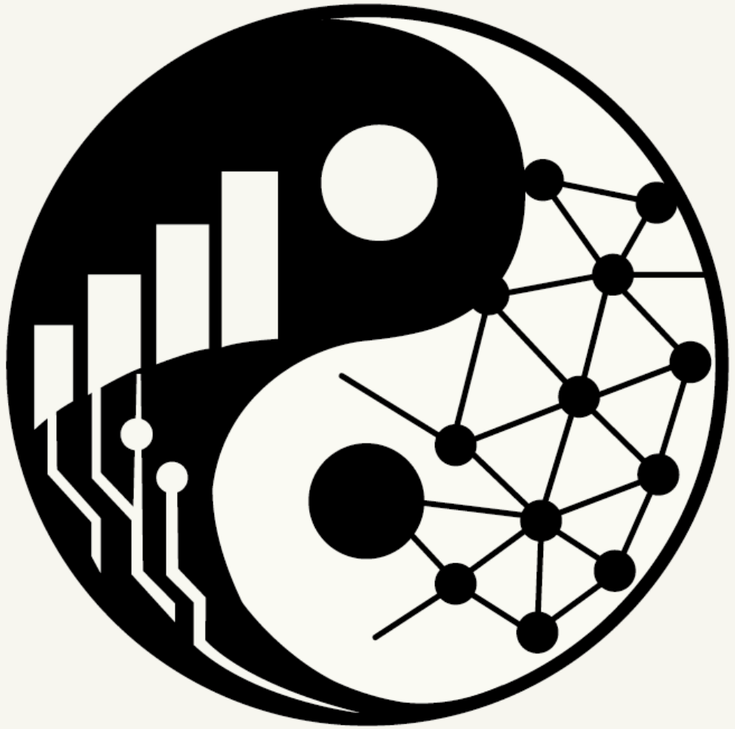
4. Identity (Z-axis / Container)

What is the spatial boundary?
After the contents are replaced, is id(C) invariant?

The judgment is not an intuitive guess; it runs through all three layers. Type Logic provides a repeatable, reusable toolset for ontological engineering.

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LonGreat Vision: to create intelligent life that understands the essence of the world



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"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence — this is the eternal path to AGI that LonGreat pursues.

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